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Accredited by NBA and with 'A+' grade by NAAC
Samayapuram, Tiruchirappalli - 621 112, Tamilnadu, India.
Incubating Minds, Catalyzing Careers

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

20CS8504 & PROJECT WORK

END SEMESTER PRACTICAL EXAMINATION- APRIL- 2026

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Session: AN

AI-Based Real Time Emotion Detection And Personalized Music And Activity Recommendation System

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OBJECTIVE OF THE PROJECT

1.To build an AI-based emotion detection system that identifies user mood using facial expressions and text analysis in real time.

2.To develop an intelligent AI recommendation engine that suggests personalized music and wellness activities such as games and books based on detected emotions.

3.To implement AI-driven emotion analytics to track mood patterns and continuously improve user experience and emotional well-being.

INTRODUCTION

1.Emotional well-being plays a vital role in human health, productivity, and overall quality of life in today's digital lifestyle.

2.Recent advancements in Artificial Intelligence have enabled machines to recognize human emotions through facial expressions and textual inputs with high accuracy.

3.Traditional music and wellness applications require manual mood selection and lack real-time emotional understanding.

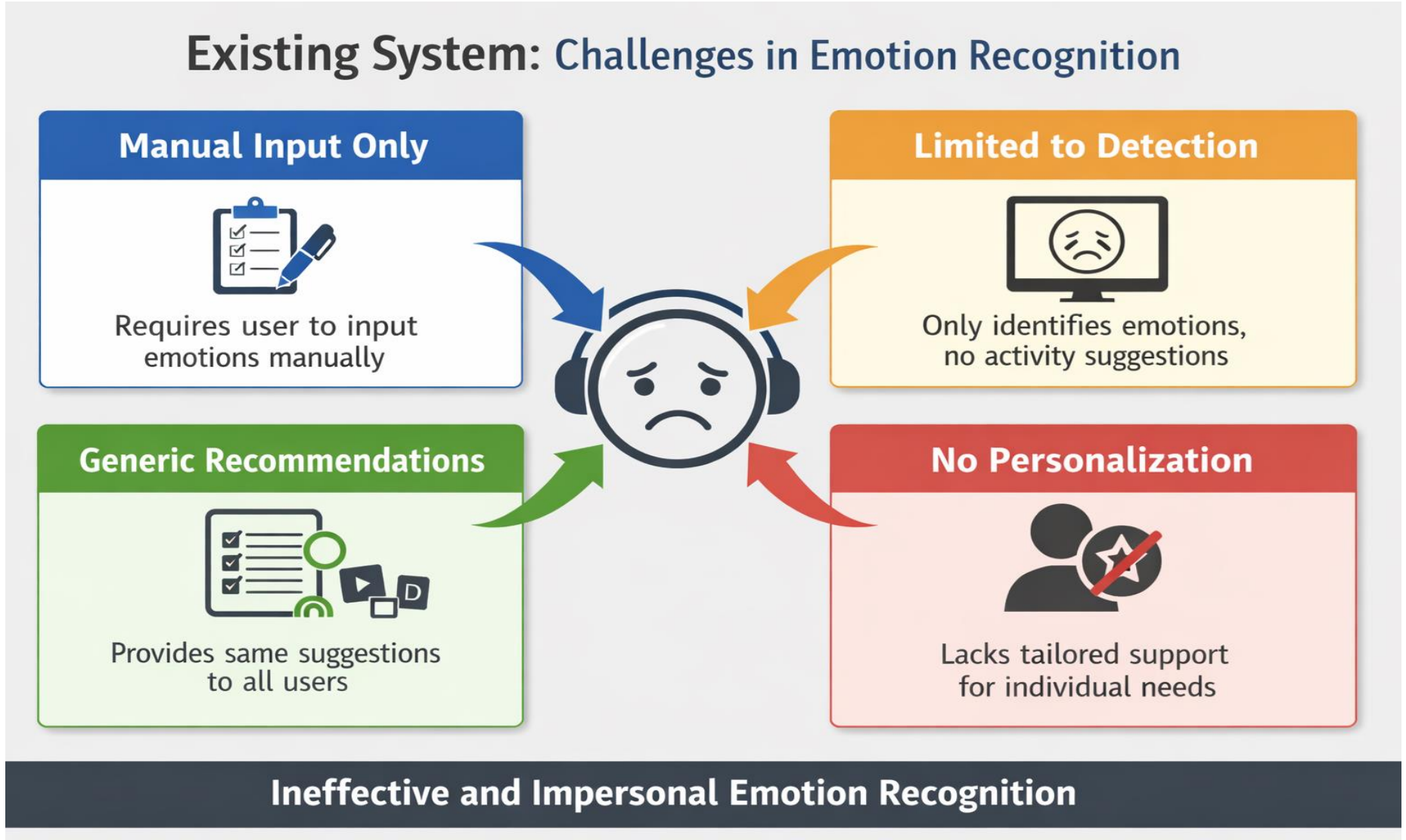
4.An AI-based emotion detection system can automatically analyze user emotions and provide personalized recommendations for relaxation, entertainment, and mental support.

5.This project integrates emotion recognition with intelligent music and wellness activity recommendations to create a smart, user-friendly emotional wellness platform.

LITERATURE SURVEY

TITLE OF PAPER	AUTHOR(S)	PAPER GIST	TECHNOLOGY USED
Smart Emotion-Based Activity Recommendation System	Sindhiya R, Akash A, Eswar M	Personalized activity suggestion	DeepFace+CNN,OpenCV, Python,HTML,CSS,JS
Music Recommendation System Based on Facial Emotion Recognition	Kreesha Iyer, Neha Grandhi, Bhagyashree Birje, Priyanka Verma	Expression-driven song selection	Convolutional Neural Networks (CNN)
Emotion Based Music Recommendation System	N. Satyanandam, Shravani Cheripally, Veldi Sriya	Mood adaptive playlist generation	CNN,Pygame,Tkinter
Intelligent Music Recommendation System on Emotion Recognition	Chaithra P, Dr Leena Giri G	Advanced emotion classification model	CNN + ResNet
AI-Based Emotion Recognition and Personalized Music	R. Sharma, K. Verma, A. Singh	Therapeutic audio personalization	Deep Learning CNN, Facial Expression Analysis

EXISTING SYSTEM ARCHITECTURE



PROBLEM IDENTIFIED

1.Many people experience emotional stress, sadness, or anxiety but do not receive immediate support or suggestions to improve their mood.

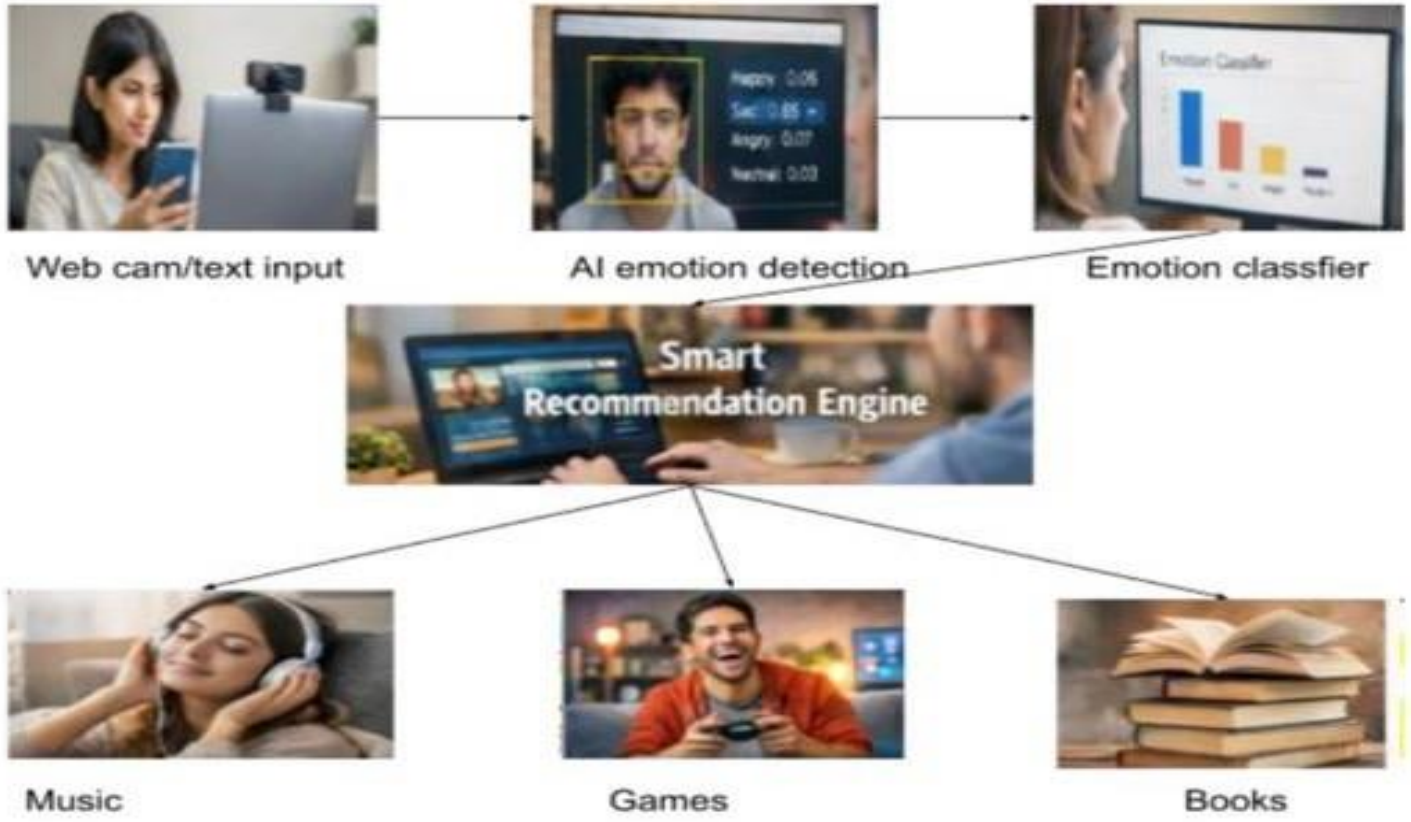
2.Traditional systems cannot automatically detect a user's emotional state in real time using facial expressions.

3.Most entertainment platforms recommend music or activities based only on user preferences, not on their current emotions.

4.Without emotion-aware systems, users may struggle to find suitable activities that help improve their mental well-being.

5.Therefore, an intelligent system is needed that can detect emotions through a webcam and provide personalized music, games, and book recommendations to enhance the user's emotional state.

PROPOSED SYSTEM ARCHITECTURE



SOFTWARE AND HARDWARE REQUIREMENTS

HARDWARE

1. System/Laptop
2. Web Camera
3. Internet Connection

SOFTWARE

1. Operating System
2. Python 3.8+
3. AI & ML Libraries(OpenCV (face detection) NumPy, Pandas)
4. NLP (for text emotion)
5. Frontend(HTML, CSS, JavaScript)
6. Backend (python)

MODULES

1. Emotion Detection Module (Face + Text)

2.Emotion Classification Module

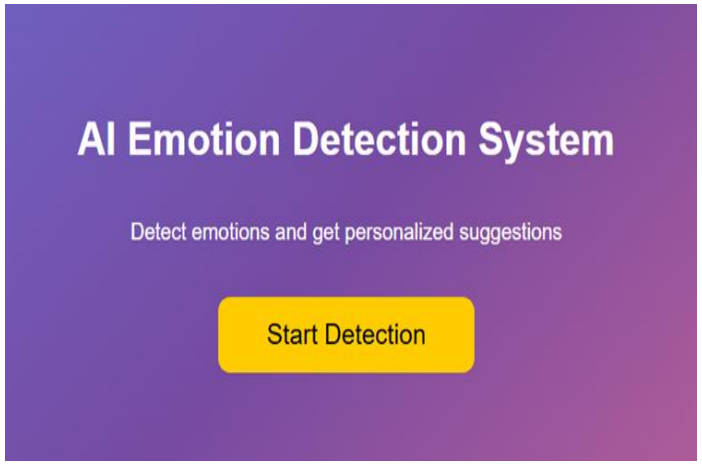
3. Wellness Activity Module

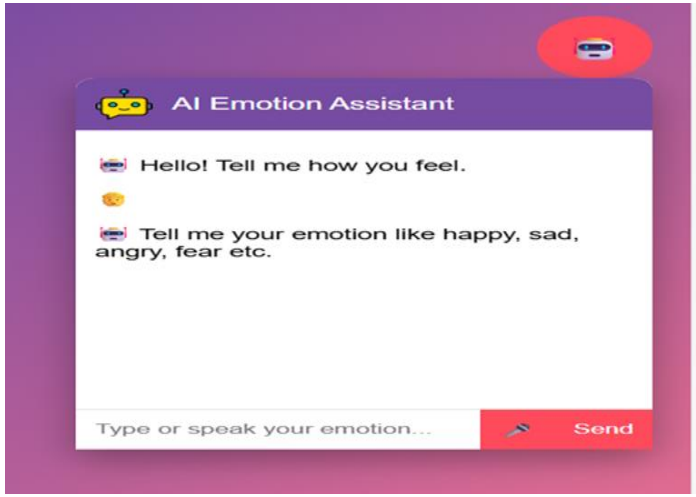
4. Smart Recommendation Engine

5. User Data & Analytics Module

Emotion Detection Module (Face + Text)

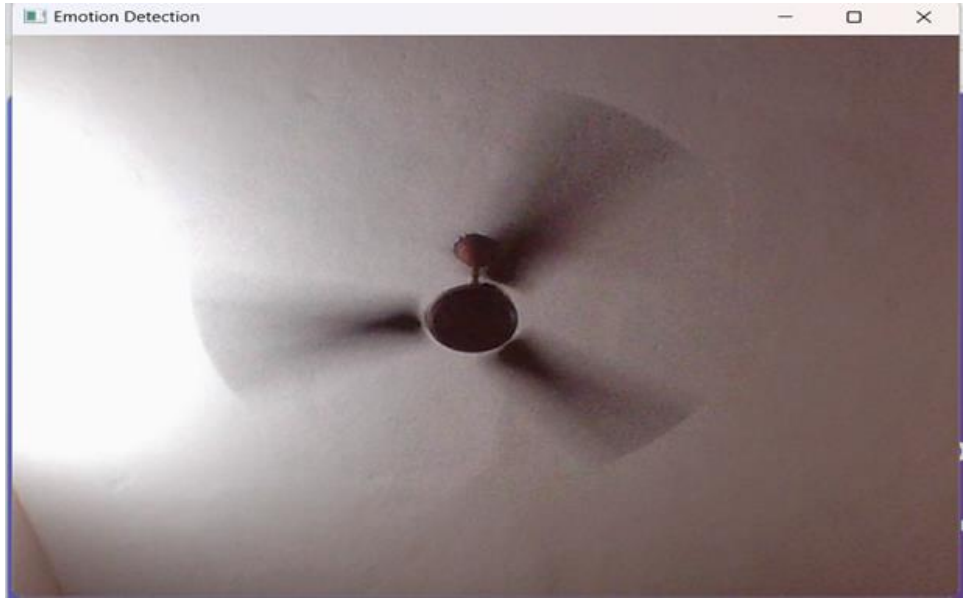
- Captures facial expressions through a camera and analyzes text inputs (chat/messages) using NLP techniques.
- Extracts emotional features such as smile patterns, eye movement, tone, and sentiment polarity.





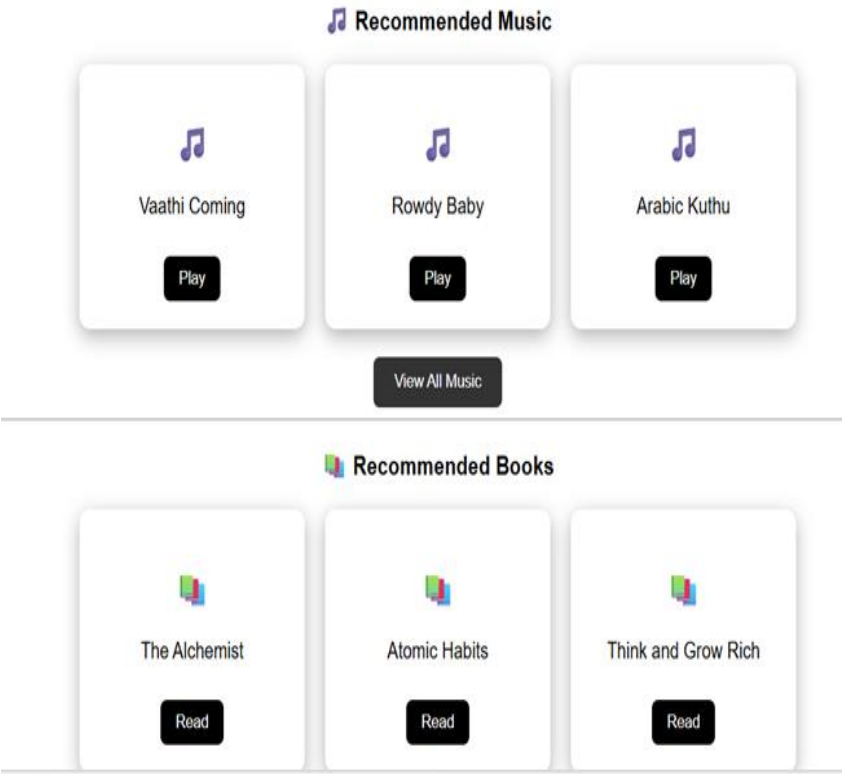
EMOTION CLASSIFICATION MODULE

- Uses machine learning / deep learning models to classify detected features into emotions like happy, sad, angry, stressed, calm, etc.
- Improves accuracy through training datasets and continuous learning from user interactions.



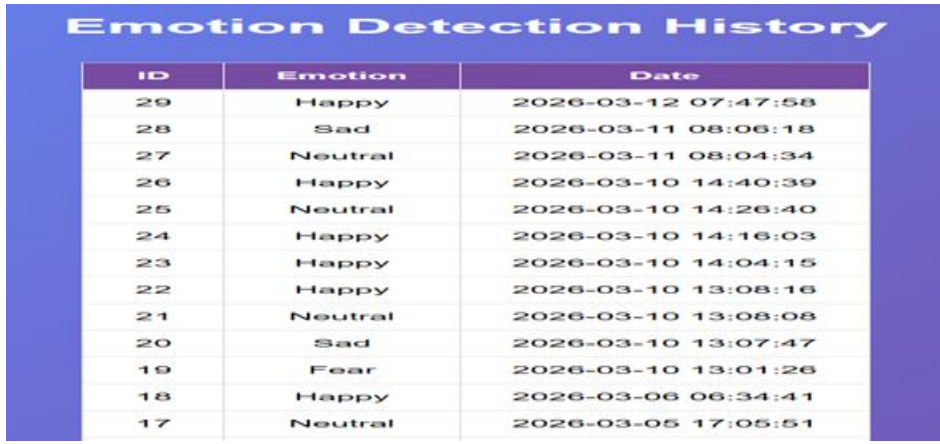
WELLNESS ACTIVITY MODULE

- Provides mood-improving activities such as listening musics, playing games and reading books.
- Tracks completion and effectiveness to recommend better coping strategies over time.



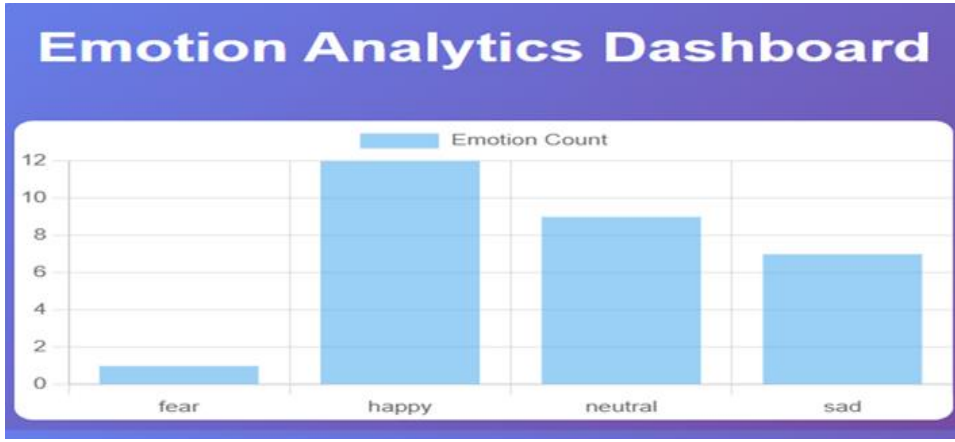
SMART RECOMMENDATION ENGINE

- Suggests suitable music playlists and activities based on the user's current emotional state.
- Applies personalization algorithms considering user preferences, history, and emotional trends.



Emotion Detection History

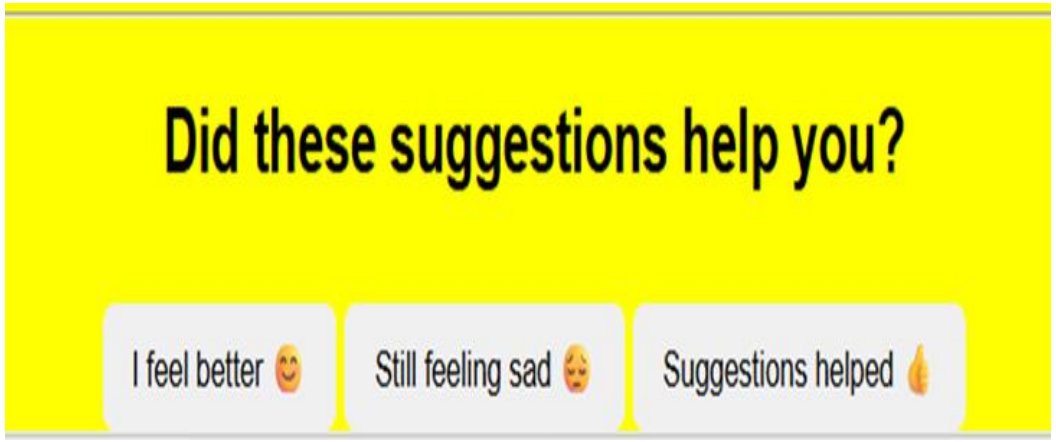
ID	Emotion	Date
29	Happy	2026-03-12 07:47:58
28	Sad	2026-03-11 08:06:18
27	Neutral	2026-03-11 08:04:34
26	Happy	2026-03-10 14:40:39
25	Neutral	2026-03-10 14:26:40
24	Happy	2026-03-10 14:16:03
23	Happy	2026-03-10 14:04:15
22	Happy	2026-03-10 13:08:16
21	Neutral	2026-03-10 13:08:08
20	Sad	2026-03-10 13:07:47
19	Fear	2026-03-10 13:01:26
18	Happy	2026-03-06 06:34:41
17	Neutral	2026-03-05 17:05:51



USER DATA & ANALYTICS MODULE

•Stores user emotion history, activity logs, and recommendation feedback securely.

•Generates pdf reports to identify emotional patterns and mental wellness trends.



CONCLUSION

1. The proposed system successfully integrates real-time emotion detection from facial expressions and text inputs to understand the user's mental state accurately.
2. Continuous data analysis and user feedback help improve recommendation accuracy and provide long-term emotional trend insights.
3. 3.Overall, the project demonstrates how AI-driven affective computing can be effectively applied to promote mental wellness and create adaptive, user-centric digital experiences.

FUTURE ENHANCEMENTS

1.The system can be integrated with smart devices such as smart speakers, smart lights, or wearable devices to automatically adjust music, lighting, or activities based on the user's emotional state.

2.In the future, the system can be developed as a mobile application so users can detect emotions and receive personalized music or activity recommendations anytime using their smartphone camera.